

DOE2 keystones: knee $\times$ dV/dQ SOC0

SJ900 vs SJ1300 · 양극 동일 · 음극 상이

DOE2 핵심 2지표: knee $\times$ dV/dQ SOC0

양극 동일 · 음극 상이. **사이클 변곡점(knee)** 과 **방전 dV/dQ @SOC0(저SOC cliff)** 가 anode arm을 가르는 1차 관측량.

- SJ900 (excl Ch025): M01Ch022, M01Ch024

- SJ1300: M01Ch010, M01Ch011, M01Ch012

1) Knee (bilinear on routine SoHQ)

- **SJ900**: knee $\approx$ 320 cyc (M01Ch022=350, M01Ch024=290); severity=0.022; SoHQ@knee $\approx$ 80.8%; end $\approx$ 64.9%

- **SJ1300**: knee $\approx$ 253 cyc (M01Ch010=250, M01Ch011=260, M01Ch012=250); severity=0.0422; SoHQ@knee $\approx$ 88.8%; e

2) dV/dQ @ SOC0 (저SOC cliff intensity)

- **SJ900**: SOC0 0.093 $\rightarrow$ 0.17 ($\Delta=+0.0774$); cliff_width 4.50 $\rightarrow$ 6.25; Δ SOC0 half-cycle $\approx$ 480

- **SJ1300**: SOC0 0.113 $\rightarrow$ 0.075 ($\Delta=-0.0378$); cliff_width 4.56 $\rightarrow$ 3.45; Δ SOC0 half-cycle $\approx$ 47

교차 해석

- **SJ900**: knee 늦음($\sim$ 320) · SOC0 **후반에 커짐** (cliff 넓어짐) · Δ SOC0 절반은 knee **이후** ($\sim$ 480).

- **SJ1300**: knee 이름($\sim$ 253) · 더 급함 · SOC0는 **초반부터 줄고** cliff 좁아짐 · Δ SOC0 절반은 knee **이전** ($\sim$ 50).

- 같은 양극인데도 SOC0 궤적 부호가 반대 $\rightarrow$ **음극 쪽 저SOC 형상**이 knee 타이밍과 함께 arm을 구분.

보조 패턴 (Δ late-early, excl Ch025)

- `LLI`: SJ900 +0.0522 vs SJ1300 +0.381

- `LAM\_PE activity`: SJ900 +0.371 vs SJ1300 +0.321

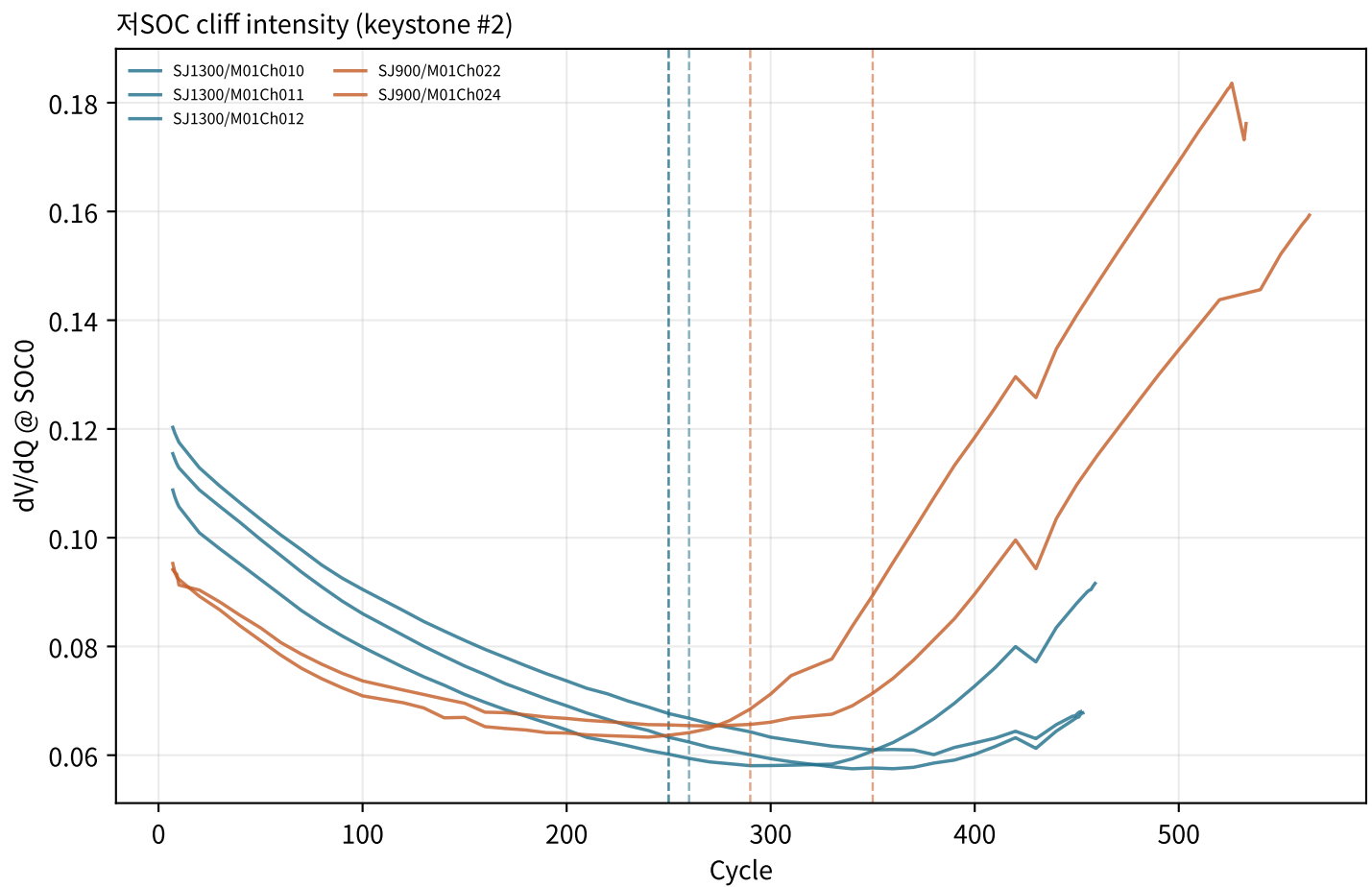
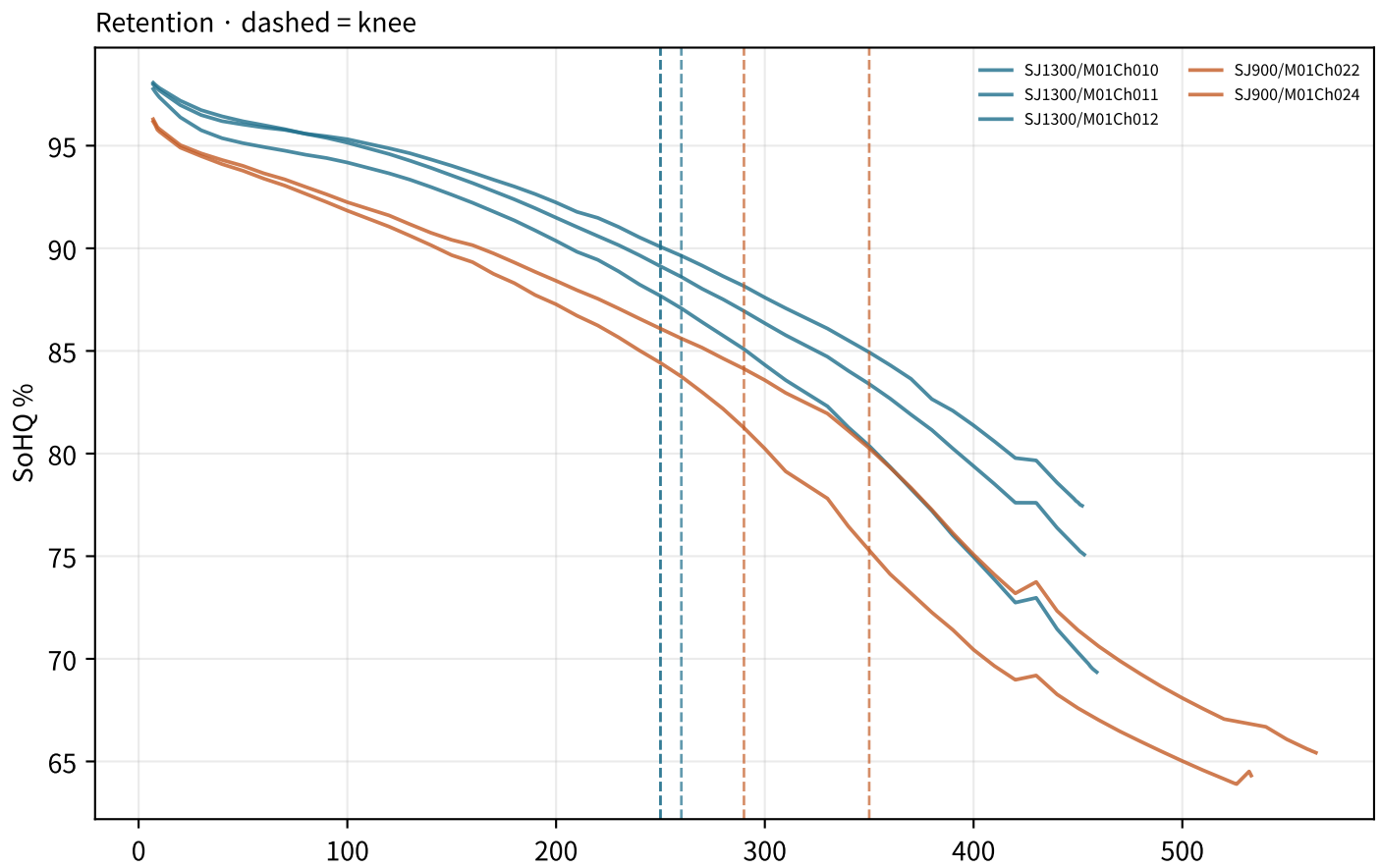
- `contact\_loss`: SJ900 +0.211 vs SJ1300 +0.183

- `si\_csign`: SJ900 +0.2 vs SJ1300 +0.0667

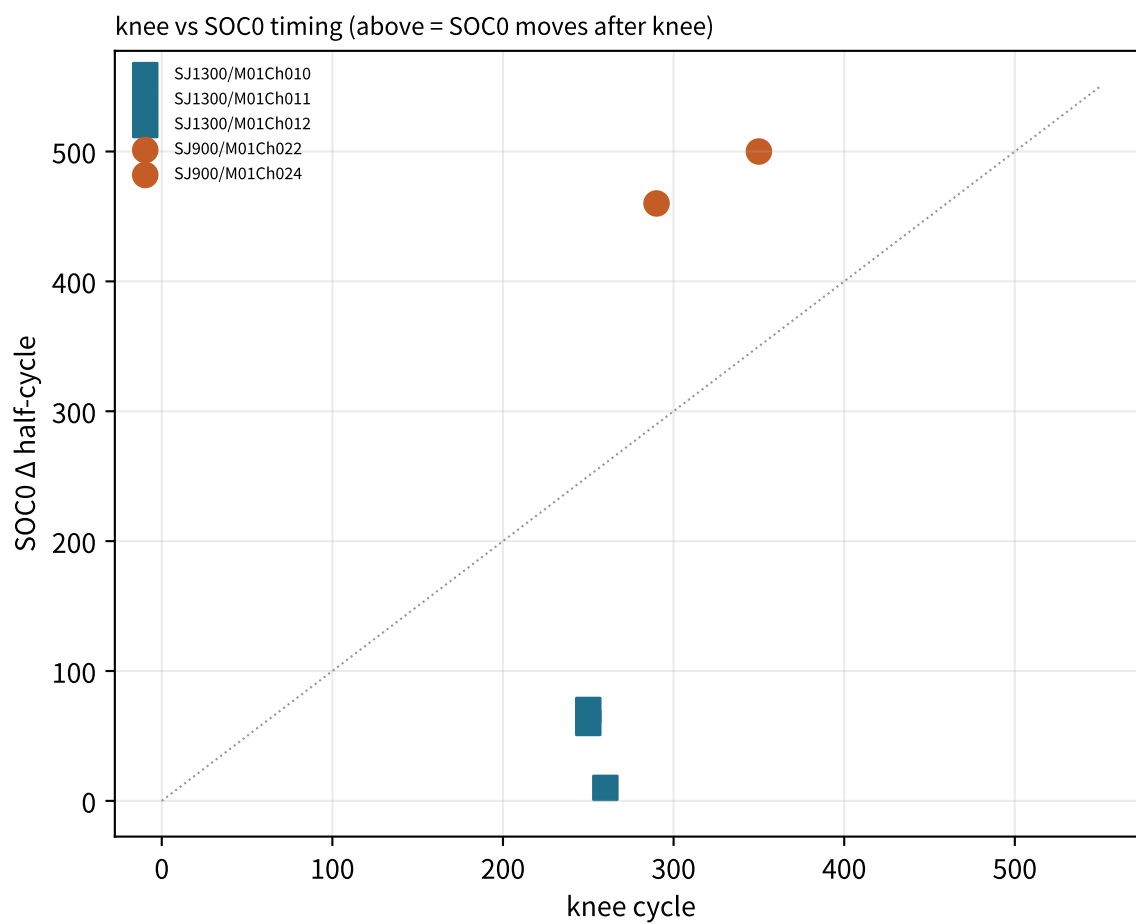
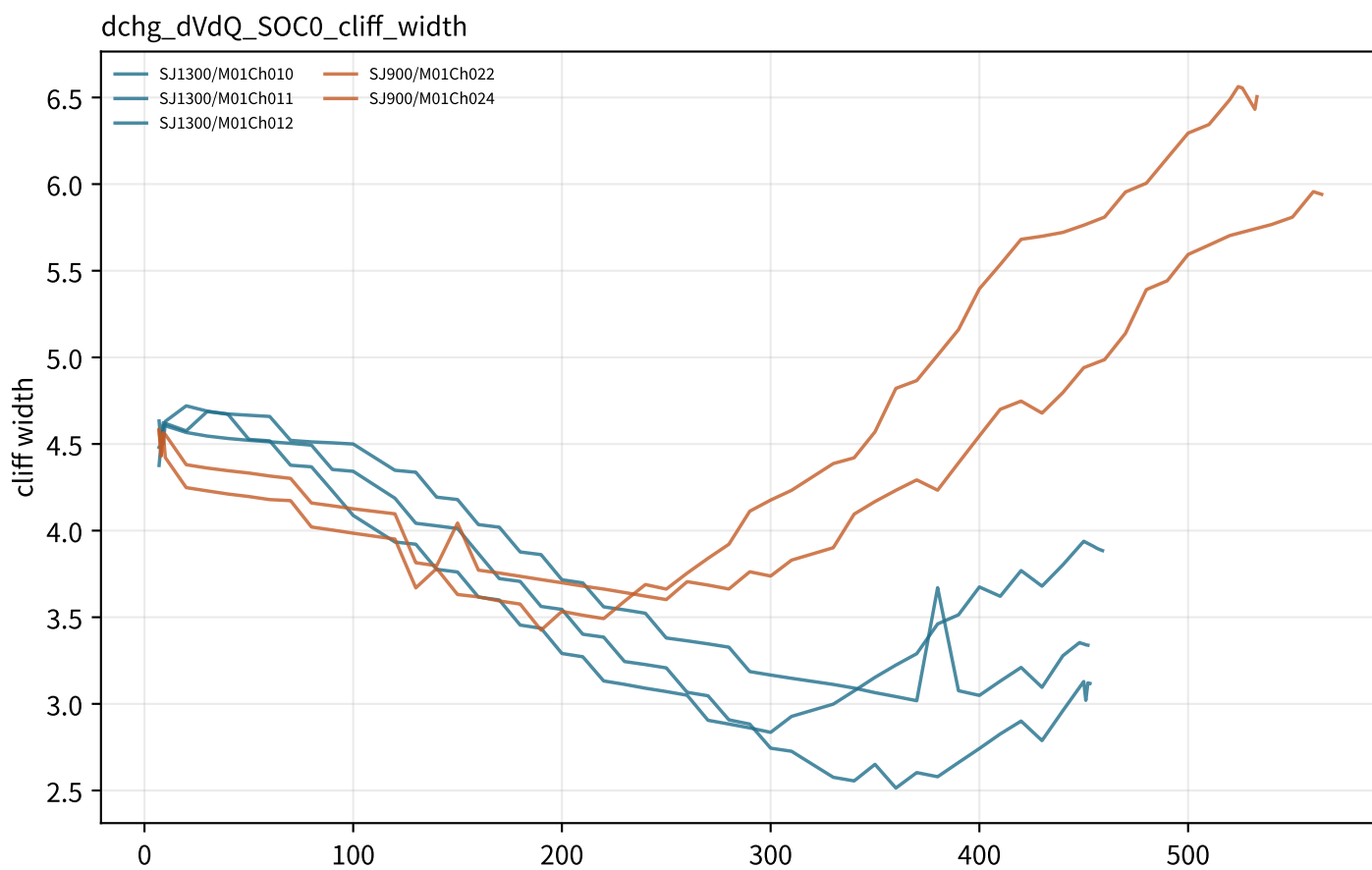
- `PE\_side`: SJ900 +0.214 vs SJ1300 +0.279

> dchg_dVdQ_SOC0 = 방전 말단(SOC $\approx$ 0) |dV/dQ|. cliff_width = mid 대비 2 $\times$ 넘는 Q폭. Ch025는 knee 평균에서 제외.

1. Keystones — SoHQ knee + dV/dQ SOC0



2. SOC0 cliff width · knee timing vs SOC0 timing



3. Supporting pattern Δ (late-early)

